Christopher Thomas Ryan (October 26, 2026)

CONTACT INFORMATION

2053 Main Mall
UBC Sauder School of Business — *Voice:* (604) 822-8435
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EMPLOYMENT

University of British Columbia, Sauder School of Business

Advisory Council Chair in Management Science	July 2022-present.
Professor	July 2024 – present.
Associate Professor (with tenure)	July 2021 – June 2024.
Assistant Professor	September 2019 – June 2021.

Harvard Business School

Visiting scholar — Academic year 2023-4

University of Chicago Booth School of Business

Associate Professor of Operations Management	July 2016 – August 2019.
Assistant Professor of Operations Management	July 2010 – June 2016.

AFFILIATIONS

Embedded Development Lab (EDLab) at Harvard Graduate School of Education

Affiliated researcher — November 2025 – present

UBC Science and Technology Studies (Faculty of Arts)

Affiliated faculty — October 2026 – present

Colibri Learning Foundation (colibrilearning.org)

Curriculum development	March 2004 – present.
Discussion leadership training	September 2006–present.

EDUCATION

University of British Columbia

Ph.D., Sauder School of Business, 2010

- Dissertation Topic: "Computing solution concepts in games with integer decisions"
- Advisor: Maurice Queyranne

B.A. (honors), Mathematics, 2005

University of Guelph

B. A. (honors), Sociology and Economics, (transferred to UBC in September 2002)

PAPER PUBLICATIONS

Published or forthcoming journal publications

(J-1) J.D. Camm and C.T. Ryan (2026). The Fifth Column Returns. *INFORMS Journal of Applied Analytics* 56(4), iii-vi.

(J-2) J. Han, C.T. Ryan, and X.T. Tong (2026). Algorithms for loot box design. *Operations Research* 74(3), 1321-1338.

(J-3) H.-S. Ahn, C.T. Ryan, J. Uichanco, and M. Zhang (2026). Valuing influence with social learning. *Manufacturing & Service Operations Management*, 28(4), 1133-1153.

(J-4) R. Chen, B. Jiang, C.T. Ryan, and N. Zhang (2026). Assortment optimization with α-similar substitutes: Insights from customer browsing patterns. *Management Science*. 72(8), 7233-7253.

(J-5) H.-S. Ahn, C.T. Ryan, J. Uichanco, and M. Zhang (2026). Certainty-equivalent pricing with dependent demand and limited price-changing opportunities. *Mathematics of Operations Research*. 51(1), 498-541.

(J-6) O. Hanguir, W. Ma, J. Han, and C.T. Ryan (2025).Optimizing for strategy diversity in the design of video games. *Mathematical Programming (Series B)* Volume 210, pages 335–376. (Preliminary version accepted at IPCO 2023).

(J-7) C.T. Ryan, L. Sheng, and X. Zhao (2025). Selling bonus actions in video games. *Management Science*, Volume 71(3), 2544–2564.

(J-8) S. Kiatsupaibul, G. Pedrielli, C.T. Ryan, R.L. Smith, and Z.B. Zabinsky (2024). Monte Carlo fictitious play for finding a Nash equilibrium of an identical interest game. *INFORMS Journal on Optimization*, Volume 6(3-4), 155–172.

(J-9) C.T. Ryan, R.L. Smith, and M.A. Epelman (2024). Minimum spanning trees in infinite graphs: Theory and Algorithms. *SIAM Journal on Optimization*, Volume 34(3), 3112–3135.

(J-10) Y.K. Che, J. Kim, F. Kojima, and C.T. Ryan (2024). "Near" Weighted Utilitarian Characterizations of Pareto Optima. *Econometrica*, Volume 92(1), 141–165.

(J-11) Y. Li, C.T. Ryan, and L. Sheng (2023). Optimal sequencing in single-player games. *Management Science*, 69(10), 6057–6075.

(J-12) C.T. Ryan and R.L. Smith (2022). A greedy algorithm for finding maximum spanning trees in infinite graphs. *Operations Research Letters*, 50(6), 655–659.

(J-13) Y. Feng, R. Caldentey, and C.T. Ryan (2022). Robust learning of consumer preferences. *Operations Research*, 70(2), 918–962.

(J-14) L. Sheng, M. Nagarajan, C.T. Ryan, Y. Cheng and C. Tong (2022). Incentivized actions in freemium games. *Manufacturing & Service Operations Management*, 24(1), 275–284.

(J-15) A. Ghate, C.T. Ryan, and R.L. Smith (2021). A simplex method for countably-infinite linear programs. *SIAM Journal on Optimization*, 31(4), 3157–3183.

(J-16) T. Dai, R. Ke, and C.T. Ryan (2021). Incentive design for marketing-operations multitasking. *Management Science*, 67(4), 2211–2230.

(J-17) C.T. Ryan and R.L. Smith (2021). Dual-based methods for solving infinite-horizon nonstationary deterministic dynamic programs. *Mathematical Programming*, 187(1), 253–285.

(J-18) X. Chen, S. He, B. Jiang, C.T. Ryan, and T. Zhang (2021). The discrete moment problem with nonconvex shape constraints. *Operations Research*, 69(1), 279–296.

(J-19) A. Basu, C.T. Ryan, and S. Sankaranarayanan (2021). Mixed-integer bilevel representability. *Mathematical Programming*, 185(1), 163–197.

(J-20) A. Basu, K. Martin, C.T. Ryan, and G. Wang (2019). Mixed-Integer Linear Representability, Disjunctions, and Chvátal Functions—Modeling Implications. *Mathematics of Operations Research*, 44(4), 1264—1285.

(J-21) R. Ke and C.T. Ryan (2018). A general solution method for moral hazard problems. *Theoretical Economics*, 13(3), 1425—1481.

(J-22) R. Ke and C.T. Ryan (2018). Monotonicity of optimal contracts without the first order approach. *Operations Research*, 66(4), 1101–1118.

(J-23) Y. Ding, D. Ge, S. He and C.T. Ryan (2018). A non-asymptotic approach to analyzing kidney exchange graphs. *Operations Research*, 66(4), 918–935.

(J-24) C.T. Ryan, R.L. Smith, and M. Epelman (2018). A simplex method for uncapacitated pure-supply infinite network flow problems. *SIAM Journal on Optimization*, 28(3), 2022–2048.

(J-25) A. Basu, K. Martin and C.T. Ryan (2017). Strong duality and sensitivity analysis in semi-infinite linear programming. *Mathematical Programming*, 161(1-2), 451–485.

(J-26) K. Martin, C.T. Ryan and M. Stern (2016). The Slater conundrum: Duality and pricing in infinite dimensional optimization. *SIAM Journal on Optimization*, 26(1), 111–138.

(J-27) A. Basu, K. Martin, and C.T. Ryan (2014). Projection: A unified approach to semi-infinite linear programs and duality in convex programming. *Mathematics of Operations Research*, 40(1), 146–170.

(J-28) A. Basu, K. Martin, and C.T. Ryan (2013). On the sufficiency of finite support duals in semi-infinite linear programs. *Operations Research Letters*, 42(1), 16–20.

(J-29) M. Köppe, C.T. Ryan, and M. Queyranne (2011). Rational generating functions and integer programming games. *Operations Research*, 59(6), 1445–1460.

(J-30) M. Köppe, M. Queyranne and C.T. Ryan (2010). Parametric integer programming algorithm for bilevel mixed integer programs. *Journal of Optimization Theory and Applications*, 146(1), 137–150.

Published peer-reviewed conference proceedings

(C-1) W. Guo, Y. Hur, T. Liang, and C.T Ryan (2022). Online learning to transport with the minimal selection principle. *COLT 2022*.

(C-2) C.T. Ryan, A. X. Jiang and K. Leyton-Brown (2010). Computing pure strategy Nash equilibria in compact symmetric games. *EC 2010*.

Under revision

(R-1) J. Han, M. Nagarajan, C.T. Ryan, and X.T. Tong. Contextual group design in online services. Major revision at *Manufacturing & Service Operations Management*.

(R-2) M. Zhang, C.T. Ryan, W. Sun, S. Subramanian and M. Ettl. Attribute-based pricing: A novel formulation and convergent algorithms. Major revision at *INFORMS Journal on Computation*.

Submitted papers

(S-1) C.T. Ryan and R.L. Smith. A geometric simplex method in infinite-dimensional spaces.

(S-2) R. Ke, C.T. Ryan, and N. Zhang. When Are Quota-Bonus Contracts Optimal? A Minimax Approach to Salesforce Compensation.

(S-3) J. Runge, C.T. Ryan. Skimming the first-penny gap: Marketing endowments in virtual worlds.

(S-4) C.T. Ryan and M. Lee. What was new in Operations Research?

(S-5) C.T. Ryan. The Science of Better Than What?

Working papers

(WP-1) Z. Ling, M. Zhang, and C.T. Ryan. On the interplay between pricing and information policies under social learning.

(WP-2) C. Lam, C.T. Ryan, and M.X. Wu. Managing the release of consumable digital goods in freemium games.

(WP-3) S. He, M. Zheng, C.T. Ryan, D. Yao. Operational transparency: Showing we are different.

(WP-4) Y. Li, C.T. Ryan, L. Sheng, and B. Wong. Optimal world design in video games.

Case studies

Published

N. Zhang, C.T. Ryan. Westown Shoe Clinic: Carrying On Del Guercio's Legacy. Ivey Publishing Case W48070, January 2026.

A. Moreno, C.T. Ryan, and T. Quinn. Jan and Jul: Outgrowing Amazon? Harvard Business School Case 626-010, November 2025.

C.T. Ryan. Faubourg: Maintaining art de vivre despite employee turnover. Ivey Publishing Case W44751, September 2025.

C.T. Ryan, K. Marsh and L. Joseph. Ezza: Scaling the nail salon. Ivey Publishing Case W44753, August 2025.

E. Paulson, C.T. Ryan, and N. Zhang. VOCEL (B): Powered by VOCEL. Harvard Business School Case 625-082, January 2025.

E. Paulson, C.T. Ryan, and N. Zhang. VOCEL (A): Democratizing Brain Science for Early Childhood Education. Harvard Business School Case 625-081, January 2025.

K. Ferriera, C.T. Ryan, and S. Mehta. ReUp Education: Can AI Help Learners Return to College? Harvard Business School Case 624-007, October 2023.

Draft available

C.T. Ryan, C. Khamsi, and Alex Sanderson. Kumwe Harvest: Evolving agricultural supply chains in Rwanda.

Grants, Honors

Social Sciences and Humanaties Research Council of Canada (SSHRC) Insight Development Grant, 2024-2025

Sauder School of Business, CPA Graduate Teaching Award, 2022

Sauder School of Business, Junior Faculty Research Award, 2020

Sauder School of Business Exploratory Research Grant, 2020-2022

Natural Sciences and Engineering Research Council of Canada (NSERC) Discovery Grant, 2020-2025

Booth School of Business, Faculty Excellence Award for teaching in the Evening and Weekend MBA program, 2018

Natural Sciences and Engineering Research Council of Canada (NSERC) Postgraduate Scholarship, 2005-2009

Shelby L. Brumelle Memorial Scholarship, 2008-2009

NSERC Undergraduate Student Research Award, 2004

Chancellor's Entrance Scholarship, University of British Columbia, 2002-2005

Canada Post Corporation Scholarship, 1998-2001

Board of Governor's Scholarship, University of Guelph, 1998-2001

Governor General Bronze Medal, 1998. Top graduating student from Penticton Secondary School.

Presentations

Plenaries

2021: INFORMS Speaker Series Webinar

Invited seminars

2026: Hong Kong University, Chinese University of Hong Kong, Hong Kong University of Science and Technology, Shanghai Jiatong (twice), Shandong University, Shanghai University of Finance and Economics, University of Texas at Austin, IESE Business School (invited)

2025: Shanghai Jiatong University, Shanghai University of Finance and Economics, Berkeley (Haas), Harvard Graduate School of Education (Embedded Development Lab)

2024: Google Research—Cambridge, UT Austin (McCoombs), Harvard Business School TOM Unit, University of Florida ISOM workshop, Harvard Business School Case Writing Group, Columbia (DRO), Columbia Economics (PER short course), CMU (Tepper), Western (Ivey), Johns Hopkins (Carey), Stanford (MS&E)

2023: University of Virginia (Darden), University College London, London Business School, MIT (DSL)

2022: Penn State, Virginia Tech INFORMS student chapter

2021: Arizona State University, University of British Columbia (Institute for Applied Mathematics), Dartmouth (Tuck), Shanghai Jiaotong, Xian Jiaotong-Liverpool, NYU Shanghai, National University of Singapore, Virginia Tech INFORMS student chapter, University of Illinois-Chicago, Boston College

2020: CMU Tepper, University of British Columbia (Vancouver School of Economics), University of British Columbia (Sauder)

2019: Washington University in Saint Louis, UT Austin, University of Florida, Chinese Academy of Science, Central University of Finance and Economics, Beijing Normal University, Rice, Columbia (joint IEOR-DRO)

2018: University of Toronto (Rotman), Johns Hopkins (Carey), Harvard Business School, Chinese Academy of Science (Institute for Computational Mathematics), Hong Kong Baptist University (Economics), Minzu University, University of Wisconsin-Madison (Business School and Industrial Engineering), Duke Fuqua, University of Michigan (Ross and IOE), UCLA (Anderson), MIT (IDSS), University of Virginia (Darden), Cornell (Johnson), Wake Forest (Business School), University of British Columbia (Sauder), University of Alberta (Business School)

2017: University of Southern California (ISE), University of British Columbia (Sauder), Simon Fraser University (Mathematics), University of Chicago (Booth)

2016: University of Kansas, University of Washington (IE), University of Waterloo (Management Sciences), University of Toronto (IE), University of British Columbia

2015: Chinese University of Hong Kong (Economics), University of Alberta

2014: New York University, University of California-Irvine, Northwestern University, University of Michigan-Ann Arbor (IOE), Georgia Tech (ISyE), CMU Tepper, Shanghai University of Finance and Economics, Massachusetts Institute of Technology, Haverford College (Mathematics), University of Indiana-Bloomington (Telecommunications), University of Minnesota (ISyE), University of British Columbia-Okanagan (Mathematics)

2012: University of British Columbia, Universidad de Chile, Chinese Academy of Science

2010: University of Southern California, University of Chicago, New York University, MIT

2009: Chinese Academy of Science, Peking University, University of California-Davis

2008: University of British Columbia, Simon Fraser University, University of Washington, University of Magdeburg

Invited conference presentations

2027: POMS Annual meeting (invited), Utah Winter Operations Conference (invited)

2025: POMS, INFORMS Annual meeting

2024: INFORMS Annual meeting

2023: POMS, INFORMS Annual Meeting, SIAM Conference on Optimization (SIOPT), IPCO

2022: Mixed Integer Programming workshop (MIP)

2019: POMS, INFORMS Annual Meeting

2018: INFORMS Optimization, POMS, International Symposium of Mathematical Programming (ISMP), INFORMS Annual Meeting

2017: SIAM Conference on Optimization (SIOPT), INFORMS Annual Meeting

2016: International Conference on Continuous Optimization (ICCOPT), INFORMS Annual Meeting

2015: INFORMS Annual Meeting, International Symposium of Mathematical Programming (ISMP)

2014: INFORMS Annual Meeting, Manufacturing and Services Operations Management (MSOM), SIAM Conference on Optimization (SIOPT)

2012: INFORMS Annual Meeting, ISMP

2010: Behavioral and Quantitative Game Theory, INFORMS Annual Meeting

2009: ISMP, American Mathematical Society (AMS) Spring Western Section Meeting, San Francisco, 25–26 April 2009.

2008: American Mathematical Society (AMS) Fall Western Section Meeting

Service

Editorial positions
Incoming chair of the Magazine Editorial Advisory Board of INFORMS (overseeing *OR/MS Today*, *Analytics*, and *OR/MS Tomorrow*)

Associate editor at *Management Science* (2020-2023)

Associate editor at *Operations Research Letters* (2019-2022)
Associate editor at *Management Science* (2020-2023)

Program Committees
ACM Conference on Economics and Computation 2018 (ACM-EC'18)

Workshop on Operations of People-Centric Systems at EC'21 (ACM-EC'21)

Ad-hoc reviewer
Operations Research, Management Science, Manufacturing & Service Operations Management, Production and Operations Management, Mathematics of Operations Research, Econometrica, Journal of Mathematical Economics, Games and Economic Behavior, Mathematical Programming, SIAM Journal on Optimization, INFORMS Journal of Computing, European Journal of Operational Research, International Journal of Game Theory, Networks, Journal of Mathematical Economics, Operations Research Letters, Economics Letters, Journal of Combinatorial Optimization, Workshop in Networks and Economics (WINE), Optimization Letters, Healthcare Management Science, IISE Transactions

INFORMS Early Career Teachers Network (2023)

NSF panelist (2015)

NSERC Discovery Grant reviewer (2017, 2018, 2021)

PhD students (advisor/co-advisor role)
Jiangze Han (UBC, current academic position: Chinese University of Hong Kong), Nanxi Zhang (SUFE, advisor during one year visit to UBC, current academic position: Ivey School of Business, Assistant Professor), Meng Zheng (SUFE, advisor during one year visit to UBC), Navid Jaberi (UBC, interdisciplinary studies with sociology), Xinyue Cheng (UBC Sauder), Ziyan He (UBC Sauder), Yixuan Xiong (SUFE, advisor during one year visit to UBC)

PhD committee member
Vishal Ahuja (Booth, current academic position: Southern Methodist University, Associate Professor), Xiao Wu (Booth, current academic position: MIT Data Science Lab), Angelo Mancini (Booth), Matt Stern (Booth), Lifei Sheng (UBC, current academic position: University of Houston, Clear Lake, Assistant Professor), Yifan Feng (Booth, current academic position: National University of Singapore, Assistant Professor), Mengzhenyu Zhang (UMichigan Ross, current academic position: UCL, Assistant Professor), Meichun Lin (UBC Sauder, current academic position: Singapore Management University, Assistant Professor), Zi (Elaine) Ling (Booth), Zhuyu Liu (UBC Sauder)

Academic mentor to undergraduate students, with most recent academic position indicated
Runshan Fu (NYU Stern, assistant professor of Marketing), Mengzhenyu Zhang (UCL, Assistant Professor of Operations Management), Jiding Zhang (Arizona State University, Assistant Professor of MIS), Teng Zhang (graduated with PhD from Stanford MS&E), Wenjia Ba (UBC Sauder, Assistant Professor of Marketing), Tongxin Zhou (Arizona State Univeristy, Assistant Professor of MIS), Ren Yi (graduated at Columbia University in IEOR), Siyue Liu (PhD student at CMU Tepper in Operations Research), Ethan Che (graduated with a PhD student from Columbia Business School), Zhejun Xi (PhD student at University of Toronto)

Conference organization
INFORMS Annual Meeting (Organizing Committee, Committee's Choice Co-Chair, Seattle, 2024)

INFORMS-CORS International Conference (Program Committee Co-Chair, Vancouver, 2022)

Mixed Integer Programming Workshop (local committee, hosted at Gleacher Center, University of Chicago, 2015)